Bacterial Culture

How we grow bacteria in the laboratory — and what their growth tells us

1. Bacteria are not only villains!

THINK AND WRITE (5 MINUTES)
Write down three positive things that bacteria do for you. Compare lists with a neighbour.

Helpful job	Example
Digestion and health	Gut bacteria help digest food, make some vitamins (like K and B12) and crowd out harmful germs.
Food	*Lactobacillus* turns milk into curd (dahi) and yoghurt; bacteria also help make cheese, idli and dosa batter, and pickles.
Farming	*Rhizobium* in the roots of pulses turns nitrogen from the air into a form plants can use.
Recycling	Decomposer bacteria break down dead plants and animals and return nutrients to the soil.
Medicine and industry	Engineered *E. coli* makes human insulin; some bacteria produce antibiotics; others clean up oil spills.

2. The birth of bacteriology

When	Who	What happened
1670s	Antonie van Leeuwenhoek	First to see bacteria, through his own hand-made microscopes.
1860s	Louis Pasteur	Showed that microbes cause fermentation and spoilage and do not appear from nothing.
1870s–80s	Robert Koch	Proved that specific bacteria cause specific diseases (anthrax, tuberculosis, cholera) and grew them as pure cultures.
1881–82	Fanny Hesse (and Walther Hesse)	Fanny Hesse suggested agar, which she used to set jellies, as a solid base for culture media. It replaced gelatin, which melted at body temperature.
1887	Julius Petri	Invented the flat covered dish we still call the Petri dish.

The period from the mid-1800s to the early 1900s is called the "Golden Age" of microbiology. The techniques of Koch, Pasteur and their colleagues are still in everyday use. Most involve isolating one kind of bacterium from a natural source (a sick person, soil, water) and growing it alone in a pure culture, so it can be studied.

3. What is a bacterial culture?

A bacterial culture is a method of letting bacterial cells multiply in or on a culture medium under carefully controlled conditions. The best conditions depend on the bacterium you want to grow.

Oxygen: aerobic and anaerobic bacteria

Type	Oxygen	Where you might find it
Obligate aerobe	Needs oxygen to grow	Skin surface, upper airways, *Mycobacterium tuberculosis* in the lungs
Obligate anaerobe	Cannot grow (and may be killed) in oxygen	Deep wounds, abscesses, gut, deep mud and ocean sediments
Facultative anaerobe	Grows with or without oxygen; switches between respiration and fermentation	*E. coli* — the usual lab bacterium
Microaerophile	Needs a little oxygen, but less than in air	*Helicobacter pylori* in the stomach

To grow anaerobes, microbiologists remove oxygen using sealed jars or chambers.

Other conditions

- **Temperature:** most bacteria that live with humans grow best near 37 °C (*mesophiles*); others prefer cold (*psychrophiles*) or hot places (*thermophiles*).
- **pH:** most prefer near-neutral pH (about 7).
- **Moisture:** all need water. In an incubator, plates can dry out, so the agar must hold enough water.

4. Feeding bacteria: culture media

Many bacteria are heterotrophs: they get their carbon and energy from organic food. This is true of disease-causing bacteria and most lab bacteria. Some need extra growth factors such as vitamins — "picky eaters". A culture medium must supply everything the cell needs to grow:

Need	Why it matters	Typical source in a medium
Water	Dissolves nutrients and takes part in many reactions	Distilled water

Need	Why it matters	Typical source in a medium
Carbon	The most common element in the cell: lipids, carbohydrates, proteins, nucleic acids	Sugars, alcohols, amino acids (organic); CO_2 for some bacteria (inorganic)
Nitrogen	Needed to build proteins and nucleic acids	Peptone, tryptone (protein breakdown products); nitrates
Mineral salts	Phosphate, sulphate, magnesium, calcium and others for enzymes and structures	Potassium phosphate, sodium sulphate...
Growth factors	Vitamins and other things the cell cannot make	Yeast extract, blood

In the early days Koch, Pasteur and colleagues used cow brains, potatoes and hay. Today media come as ready-made powders: you weigh the right amount, add water and mix. The medium is sterilised in an autoclave (steam under pressure, at 121 °C), and used when it has cooled.

Liquid and solid media

- **Broth (liquid medium):** nutrients dissolved in water. Bacteria grow exponentially until nutrients run out or waste products build up. Good for growing large numbers of cells.
- **Solid medium (agar plate):** made by adding 1–2 % **agar**, a jelly from seaweed (a mixture of agarose and agaropectin) that very few bacteria can digest. The hot, sterilised mixture is cooled to about 45 °C, poured into Petri dishes and left to set. Single cells land on the surface and grow into separate **colonies**. Each colony starts from one **colony-forming unit (CFU)** — often one cell — so all the cells in a colony are assumed to be identical.

Special-purpose media

Type	Idea	Examples
General / nutrient	Feeds most bacteria	Nutrient agar, LB agar
Enriched	Extra nutrients for fussy bacteria	Blood agar, chocolate agar
Selective	Contains something that stops unwanted bacteria so only the target grows	MacConkey agar (Gram-negative only), TCBS agar (*Vibrio*), SS and Hektoen agar (*Salmonella*, *Shigella*)
Differential	Shows a visible difference between kinds of bacteria	Blood agar (clear zones where red cells are destroyed), MacConkey (pink colonies ferment lactose)
Testing	Standard medium for antibiotic tests	Mueller–Hinton agar

Selective media and antibiotics

A selective medium is a base medium plus an inhibitor — an antibiotic, a dye or a chemical — that suppresses the unwanted bacteria. Antibiotics are the most common inhibitors because their effects are well known:

- **Antibacterial** antibiotics: vancomycin, penicillin G and bacitracin target Gram-positive bacteria; colistin and polymyxin B target Gram-negative bacteria.
- **Antifungal** antibiotics: nystatin, cycloheximide and amphotericin B stop fungi and yeasts.
- Combining several antibiotics makes the medium even more selective.



Figure 1. Antibiotic discs on a bacterial lawn. A clear ring (a *zone of inhibition*) around a disc means the antibiotic stopped the bacteria growing; a bigger ring means a stronger effect. *Source: Class material*

Dyes can act as colour indicators, or as inhibitors. Crystal violet, malachite green and methylene blue mainly inhibit Gram-positive bacteria. For example, crystal violet is in MacConkey agar, and methylene blue in EMB agar.

5. Bacterial growth

The time a bacterial cell takes to divide into two is the generation time. It depends on the species and conditions. E. coli can double every 20 minutes in ideal conditions.

Time	Doublings	Cells (from 1)
0	0	1
1 hour	3	8
2 hours	6	64
6 hours	18	262,144
8 hours	24	about 16.8 million

WHY DOESN'T THIS GO ON FOREVER?
At that rate one cell would become about 5×10^{21} cells in a day. It never happens, because nutrients run out and waste builds up. That is why the curve levels off.

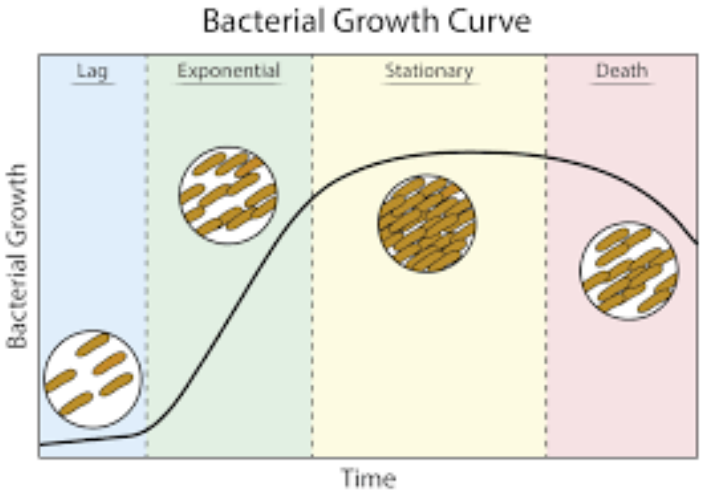


Figure 2. The bacterial growth curve in a closed system: lag, exponential (log), stationary and death phases. *Source: JoVE*

Phase	What happens
Lag	The bacteria adapt to the new medium. They do not divide yet, but they are very active: growing in size and making RNA, enzymes and growth factors.
Log (exponential)	Cells divide at a steady, fastest rate: 1 → 2 → 4 → 8. Cells are healthiest here, so researchers use log-phase cells. It ends when nutrients run low or toxic waste builds up. On a graph with a *logarithmic* y-axis this phase is a straight line, and its slope gives the growth rate.
Stationary	New cells appear at about the same rate as others die, so the number stays constant.
Death	More cells die than are born, because of lack of nutrients, toxic waste and other damage.

How do we count bacteria?

- **Plate count:** spread a diluted sample on agar and count colonies (each is one CFU).
- **Turbidity:** a cloudy culture blocks light. A spectrophotometer measures the *optical density* (OD_{600}) — more cloudy, more cells.

Check your understanding

1. Name two ways a bacterium can be useful and two ways it can be harmful.
2. Why is agar a good choice for solid media?
3. What is the difference between a selective and a differential medium?
4. Which growth phase should researchers use for experiments, and why?
5. Start with 1 *E. coli* cell and a generation time of 30 minutes. How many cells after 3 hours?

Answers are in the Quizzes and Answer Keys document.